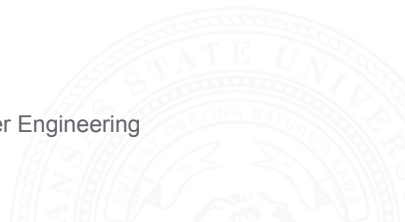


Exam 1 Practice

AC circuits, transformer models, and per-unit calculations

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Exam 1 format and timing

Friday, October 2, 2026

12:30–1:20 PM (Central) | DUE 0097

During regular class Lectures 1–14 100 points

Section	Points	Suggested time
5 multiple-choice items	20	6 min
Calculation problem 1	25	11 min
Calculation problem 2	25	9 min
Calculation problem 3	30	14 min
Checking time		10 min

Exam materials and conduct

- Bring one student-prepared formula sheet. Attach it to your exam.
- A non-programmable calculator is allowed and suggested for use.
- Work independently. No communication or access to outside resources during the exam.
- Online students may use an authorized device only for remote proctoring or submission.

Answer format and grading

- **Multiple choice:** select one answer per item. Each correct choice earns 4 points. No written explanation is required.
- **Calculations:** show equations, substitutions, and units. Report numerical results to three significant figures.
- Use RMS phasors and positive *abc* sequence. Positive load P , Q mean absorption. Source power means power delivered to the loads.
- Correct methods and consistent follow-through can earn partial credit.

Complete every section and reserve time to check signs, units, and per-unit bases.

Practice topics

1. **RMS phasors and complex power**
2. **Balanced three-phase circuits:** line/phase quantities and delta–Y equivalence
3. **Power-factor correction:** single-phase method, then parallel three-phase loads
4. **Transformer tests and referral:** SC test; OC test
5. **Per-unit calculations:** base quantities, normalization, and base changes
6. **Planning:** screening cost and reserve margin

PART 1

RMS phasors and complex power

Convert waveforms to phasors, then calculate power

RMS phasors and complex power

Key relations

$$v(t) = \sqrt{2}V \cos(\omega t + \theta_v) \iff \underline{V} = V \angle \theta_v$$

$$\underline{I} = I \angle \theta_i, \quad \underline{S} = \underline{V} \underline{I}^* = V I \angle (\theta_v - \theta_i) = P + jQ$$

$$P = V I \cos \phi, \quad Q = V I \sin \phi, \quad \text{pf} = \cos \phi, \quad \phi = \theta_v - \theta_i$$

Current relative to voltage	Absorbed Q	Power factor
Lags	Positive	Lagging
Leads	Negative	Leading

Phasor magnitudes are RMS values. The conjugate makes the power angle voltage minus current.

Example 1: sinusoidal voltage and current

A single-phase load has voltage and current

$$v(t) = 180\sqrt{2} \cos(377t - \pi/9) \text{ V},$$

$$i(t) = 12\sqrt{2} \cos(377t + \pi/18) \text{ A}.$$

The current enters the positive voltage terminal.

1. Express $\underline{V}$ and $\underline{I}$ as RMS phasors in degrees.
2. Find complex power $S = P + jQ$.
3. State the power factor and whether it is leading or lagging.

Solution 1: phasors and power

$$\underline{V} = 180\angle(-20^\circ) \text{ V}, \quad \underline{I} = 12\angle 10^\circ \text{ A}$$

$$\phi = -20^\circ - 10^\circ = -30^\circ$$

$$S = 180(12)\angle(-30^\circ) = 1870.6 - j1080 \text{ VA}$$

$$P = 1.87 \text{ kW}, \quad Q = -1.08 \text{ kvar}$$

$$\text{pf} = 0.866 \text{ leading}$$

Current leads voltage by 30° . The load absorbs real power and supplies reactive power.

PART 2

Balanced three-phase circuits

Line and phase quantities, delta loads, and total power

Balanced three-phase circuits

Key relations: positive *abc* sequence and RMS phasors

$$\underline{V}_{AN} = \frac{V_{AB}}{\sqrt{3}} \angle(-30^\circ)$$

$$\text{Delta: } \underline{I}_{AB} = \frac{V_{AB}}{Z_{\Delta}}, \quad \underline{I}_A = \underline{I}_{AB} - \underline{I}_{CA} = \sqrt{3} \underline{I}_{AB} \angle(-30^\circ)$$

$$\text{Equivalent Y: } Z_Y = Z_{\Delta}/3$$

$$\text{Phase/branch phasors } S_{3\phi} = 3 \underline{V}_{AN} \underline{I}_A^* = 3 \underline{V}_{AB} \underline{I}_{AB}^*$$

$$\text{Line RMS magnitudes } |S_{3\phi}| = \sqrt{3} V_{LL} I_L, \quad P_{3\phi} = \sqrt{3} V_{LL} I_L \cos \phi$$

For balanced sinusoids, $p_{3\phi} = P_{3\phi}$ is constant. Here ϕ is the load impedance angle.

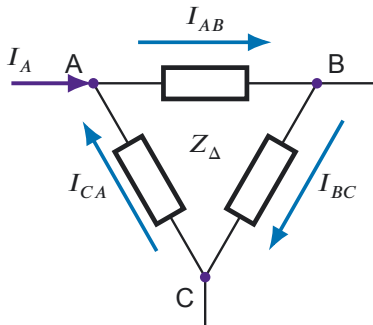
Example 2: balanced delta load

$Z_{\Delta} = 8 + j6 \, \Omega$ per branch.

$$\underline{V}_{AB} = 208 \angle 0^\circ \text{ V.}$$

Positive *abc* sequence, negligible line impedance.

1. Find V_{AN} , I_{AB} , and I_A .
2. Find total complex power and pf.
3. Check using equivalent Z_Y .



Follow the branch arrows before applying KCL at terminal A.

Solution 2: branch and line currents

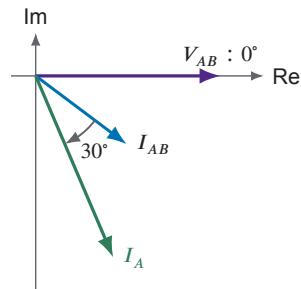
$$Z_{\Delta} = 10\angle 36.87^{\circ} \Omega$$

$$V_{AN} = 120.09\angle(-30^{\circ}) \text{ V}$$

$$I_{AB} = 20.80\angle(-36.87^{\circ}) \text{ A}$$

$$I_A = \sqrt{3}I_{AB}\angle(-30^{\circ})$$

$$I_A = 36.0\angle(-66.9^{\circ}) \text{ A}$$



V and A use separate scales

The line current lags its own phase voltage by the impedance angle.

Solution 2: power and equivalent Y

$$S_{3\phi} = 3 \frac{208^2}{8 - j6}$$

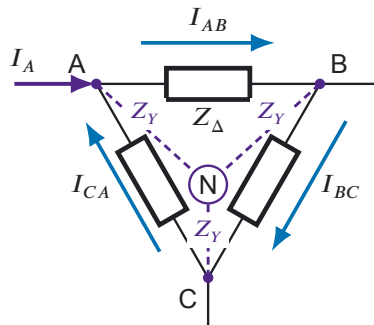
$$= 10.383 + j7.788 \text{ kVA}$$

$$|S_{3\phi}| = 12.979 \text{ kVA}$$

$$\text{pf} = 0.800 \text{ lagging}$$

$$Z_Y = \frac{8 + j6}{3} = 2.667 + j2.000 \Omega$$

$$\underline{I}_A = \frac{V_{AN}}{Z_Y} = 36.03 \angle (-66.87^\circ) \text{ A}$$



Solid: physical Δ . Dashed: equivalent Y at the same terminals.

Each Y branch has $Z_Y = Z_\Delta/3$. Line currents and total power stay the same.

PART 3

Power-factor correction

Start with the L07 method, then combine parallel-load powers

Power-factor correction: the L07 method

Key relations: unchanged load, ideal shunt capacitor

For a load $P + jQ_1$ and a **lagging** target power factor,

$$Q_{\text{target}} = P \tan(\cos^{-1} \text{pf}_{\text{target}}).$$

$$Q_{C,\text{abs}} = Q_{\text{target}} - Q_1 < 0, \quad S_C = jQ_{C,\text{abs}}.$$

Positive capacitor rating: $Q_C = |Q_{C,\text{abs}}| = Q_1 - Q_{\text{target}}.$

$$\text{Single phase: } C = \frac{Q_C}{2\pi f V^2}, \quad |I_s| = \frac{P}{V \text{pf}_s}.$$

L07 uses signed absorbed capacitor power. The equipment kvar rating is its positive magnitude.

Example 3A: single-phase correction

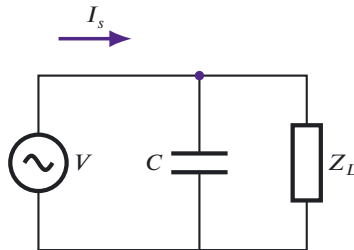
A single-phase load absorbs

$$S_L = 4.80 + j3.60 \text{ kVA.}$$

The supply is 240 V RMS at 60 Hz. Add an ideal shunt capacitor for a source pf of **0.960 lagging**.

1. Find target source Q and capacitor kvar rating.
2. Find C in μF .
3. Check source power and current before and after correction.

Use the same four steps as L07. The load and its terminal voltage remain fixed.



Same RMS voltage
across load and capacitor

Solution 3A: target power and capacitor rating

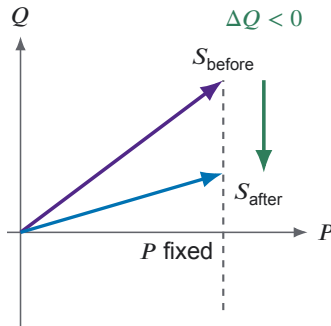
$$\text{pf}_1 = 4.80/6.00 = 0.800 \text{ lagging}$$

$$Q_{\text{target}} = 4.80 \tan(\cos^{-1} 0.960) = 1.40 \text{ kvar}$$

$$Q_{C,\text{abs}} = 1.40 - 3.60 = -2.20 \text{ kvar}$$

$$Q_C = 2.20 \text{ kvar}, \quad S_C = -j2.20 \text{ kVA}$$

$$S_s = 4.80 + j1.40 \text{ kVA}, \quad |S_s| = 5.00 \text{ kVA}$$



Compensation reduces positive source kvar while leaving real power unchanged.

Solution 3A: capacitance and current check

$$C = \frac{2200}{2\pi(60)(240)^2} = \boxed{101 \mu\text{F}}$$

$$|I_{s,1}| = \frac{6000}{240} = 25.0 \text{ A}$$

$$|I_{s,2}| = \frac{5000}{240} = \boxed{20.8 \text{ A}}$$

$$\text{pf}_{s,2} = \frac{4.80}{5.00} = \boxed{0.960 \text{ lagging}}$$

The source current decreases; the original load current stays the same.

Parallel loads: combine powers first

Combine L06 complex power, L07 correction, and L09 three-phase power

$$Q_k = \begin{cases} +P_k \tan(\cos^{-1} \text{pf}_k), & \text{lagging} \\ -P_k \tan(\cos^{-1} \text{pf}_k), & \text{leading} \end{cases}$$

$$P = \sum_k P_k, \quad Q = \sum_k Q_k, \quad |S| = \sqrt{P^2 + Q^2}$$

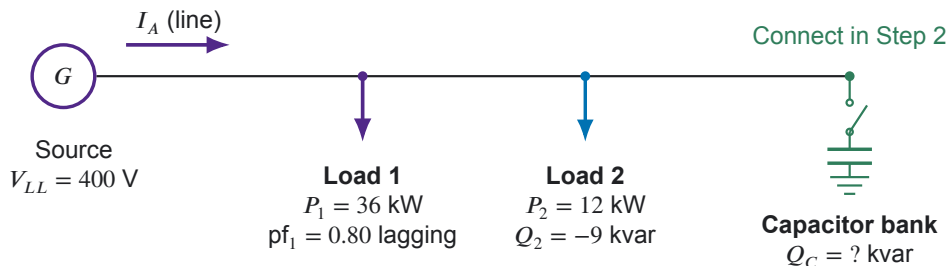
$$\text{pf}_s = \frac{P}{|S|}, \quad |I_L| = \frac{|S|}{\sqrt{3}V_{LL}}$$

$$Q_{\text{target}} = P \tan(\cos^{-1} \text{pf}_{\text{target}}), \quad Q_C = Q - Q_{\text{target}}$$

Use three-phase total powers and a lagging target. Size the bank in kvar; no bank connection is needed.

Example 3B: combine parallel loads

Balanced three-phase system (one-line diagram)



- 1. Before compensation:** find total P , Q , source power factor, and $|I_A|$.
- 2. Add the bank:** find $Q_C > 0$ (kvar) for a source power factor of **0.97 lagging**.

All powers are three-phase totals. Voltage and loads stay fixed; line impedance is negligible.

Solution 3B: combined power and compensation

$$Q_1 = 36 \tan(\cos^{-1} 0.80) = 27.0 \text{ kvar}, \quad Q_2 = -9.00 \text{ kvar}$$

$$P = 48.0 \text{ kW}, \quad Q = 18.0 \text{ kvar}, \quad |S| = \sqrt{48^2 + 18^2} = 51.3 \text{ kVA}$$

$$\text{pf}_s = \boxed{0.936 \text{ lagging}}, \quad |I_A| = \frac{|S|}{\sqrt{3}(400)} = \boxed{74.0 \text{ A}}$$

$$Q_{\text{target}} = 48 \tan(\cos^{-1} 0.97) = 12.0299 \text{ kvar}$$

$$\boxed{Q_C = 18 - 12.0299 = 5.97 \text{ kvar}}$$

Keep guard digits until the final answer. The capacitor absorbs negative kvar; its rating is positive.

PART 4

Transformer tests and referral

SC test: series impedance; OC test: excitation branch

Ideal transformer ratios

Key relations from L10: voltage, current, and referred impedance

$$a = \frac{N_H}{N_L} = \frac{|V_H|}{|V_L|}, \quad \boxed{|I_H| = \frac{|I_L|}{a}}, \quad Z_H = a^2 Z_L.$$

Quick example

For an ideal transformer with $a = 6.00$ and $|I_L| = 42.0$ A,

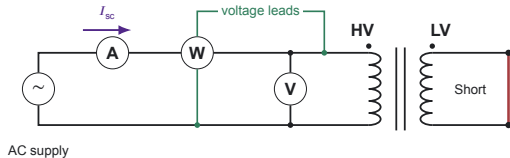
$$\boxed{|I_H| = 42.0/6.00 = 7.00 \text{ A}}.$$

The higher-voltage winding carries the lower current for the same apparent power.

For the test problem, obtain a from rated winding voltages, not from the applied SC voltage.

Transformer tests and referral

SC test: energize HV (H); short LV (L)



A: I_{SC} (RMS)

V: V_{SC} (RMS)

W: P_{SC} (real power)

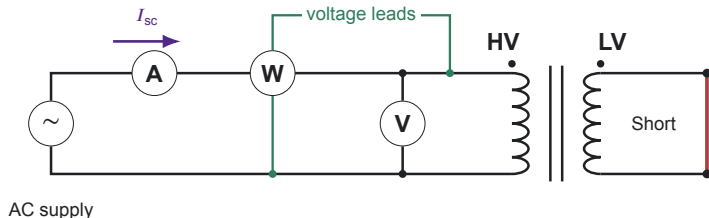
$$|Z_{eq,H}| = \frac{V_{SC}}{I_{SC}}, \quad R_{eq,H} = \frac{P_{SC}}{I_{SC}^2}, \quad X_{eq,H} = \sqrt{|Z_{eq,H}|^2 - R_{eq,H}^2}$$

$$a = \frac{N_H}{N_L} = \frac{V_{H, rated}}{V_{L, rated}}, \quad Z_{eq,L} = \frac{Z_{eq,H}}{a^2}$$

At reduced test voltage, neglect excitation; the parameters refer to H.

Example 4A: SC test and referral

Single-phase nameplate: 30 kVA, 1500/150 V (HV/LV).



Ammeter (A) **Voltmeter (V)** **Wattmeter (W)**

$$I_{SC} = 20 \text{ A}$$

$$V_{SC} = 75 \text{ V}$$

$$P_{SC} = 600 \text{ W}$$

1. Find $R_{eq,H}$, $X_{eq,H}$, and $Z_{eq,H}$.
2. Refer the equivalent series impedance to L.

HV is energized; LV is shorted. Use the approximate series model.

Solution 4A: impedance on both sides

$$|Z_{\text{eq},H}| = 75/20 = 3.75 \, \Omega$$

$$R_{\text{eq},H} = 600/20^2 = 1.50 \, \Omega$$

$$X_{\text{eq},H} = \sqrt{3.75^2 - 1.50^2} = 3.4369 \, \Omega$$

$$Z_{\text{eq},H} = 1.50 + j3.4369 \, \Omega$$

$$a = 10, \quad Z_{\text{eq},L} = \frac{1.50 + j3.4369}{100} = 0.0150 + j0.034369 \, \Omega$$

Both ohmic impedances describe the same transformer from different terminals.

Open-circuit test

Part 4 optional application: approximate excitation branch

Apply rated voltage to the test winding and leave the other winding open.

$$I_c = \frac{P_{OC}}{V_{OC}}, \quad I_m = \sqrt{I_{OC}^2 - I_c^2}$$

$$R_c = \frac{V_{OC}}{I_c} = \frac{V_{OC}^2}{P_{OC}}, \quad X_m = \frac{V_{OC}}{I_m}$$

For a test on L, refer the parameters to H using

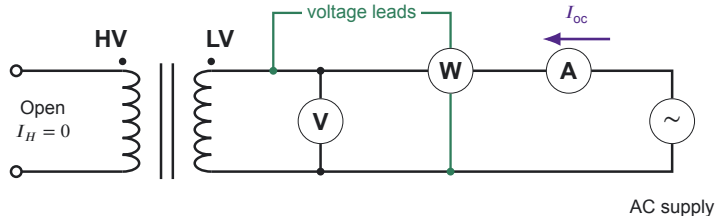
$$R_{c,H} = a^2 R_{c,L}, \quad X_{m,H} = a^2 X_{m,L}.$$

The approximate OC model assigns measured real power to core loss and neglects winding voltage drop.

Example 4B: open-circuit measurements

Optional application: energize LV (L); keep HV (H) open

Single-phase nameplate: 1200/120 V (HV/LV).



Ammeter (A) **Voltmeter (V)** **Wattmeter (W)**

$I_{OC} = 1.00 \text{ A}$ $V_{OC} = 120 \text{ V}$ $P_{OC} = 72 \text{ W}$

Find I_c , I_m , $R_{c,L}$, and $X_{m,L}$; then refer R_c and X_m to H.

The meters are on LV. The opposite winding carries no load current.

Solution 4B: excitation parameters

Optional application

$$I_c = 72/120 = 0.600 \text{ A}, \quad I_m = \sqrt{1.00^2 - 0.600^2} = 0.800 \text{ A}$$

$$R_{c,L} = 120/0.600 = 200 \text{ } \Omega, \quad X_{m,L} = 120/0.800 = 150 \text{ } \Omega$$

$$a = 10, \quad R_{c,H} = 20,000 \text{ } \Omega, \quad X_{m,H} = 15,000 \text{ } \Omega$$

OC data identify the shunt branch. SC data identify the series branch.

PART 5

Per-unit calculations

Choose bases, normalize quantities, and change reference bases

Per-unit bases

Key relations

System and chosen bases	Current base	Impedance base
Single phase: S_b, V_b	S_b/V_b	V_b^2/S_b
Balanced three phase: $S_{b,3\phi}, V_{b,LL}$	$S_{b,3\phi}/(\sqrt{3}V_{b,LL})$	$V_{b,LL}^2/S_{b,3\phi}$

$$Z_{p.u.} = \frac{Z}{Z_b}, \quad I_{p.u.} = \frac{I}{I_b}, \quad V_{p.u.} = \frac{V}{V_b}$$

Across a transformer, use the same power base and voltage bases in the turns ratio:

$$V_{b,H}/V_{b,L} = a, \quad Z_{b,H}/Z_{b,L} = a^2.$$

With compatible bases, referred physical impedance and its base change by the same factor.

Example 5A: transformer on rated bases

For the same single-phase transformer, choose

$$S_b = 30 \text{ kVA}, \quad V_{b,H} = 1500 \text{ V}, \quad V_{b,L} = 150 \text{ V}.$$

1. Find the current and impedance bases on H and L.
2. Normalize $Z_{\text{eq},H} = 1.50 + j3.4369 \Omega$.
3. Normalize $Z_{\text{eq},L} = 0.0150 + j0.034369 \Omega$ and compare.

Solution 5A: current and impedance bases

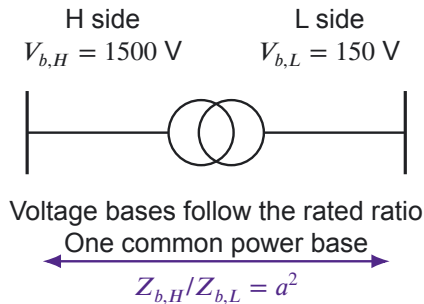
Single-phase base: 30 kVA

$$I_{b,H} = \frac{30,000}{1500} = 20 \text{ A}$$

$$I_{b,L} = \frac{30,000}{150} = 200 \text{ A}$$

$$Z_{b,H} = \frac{1500^2}{30,000} = 75 \text{ } \Omega$$

$$Z_{b,L} = \frac{150^2}{30,000} = 0.75 \text{ } \Omega$$



Current bases scale inversely with voltage. Impedance bases scale with voltage squared.

Solution 5A: normalized impedance

$$\begin{aligned}Z_{\text{eq,p.u.}} &= \frac{1.50 + j3.4369}{75} = 0.0200 + j0.0458 \\ \frac{0.0150 + j0.034369}{0.75} &= 0.0200 + j0.0458 \\ |Z_{\text{eq,p.u.}}| &= 0.0500\end{aligned}$$

Physical impedance and impedance base change by the same factor across the transformer.

Changing the per-unit base

Key relations

Physical impedance remains fixed:

$$Z = Z_{\text{p.u.,old}} Z_{b,\text{old}} = Z_{\text{p.u.,new}} Z_{b,\text{new}}.$$

$$Z_{\text{p.u.,new}} = Z_{\text{p.u.,old}} \frac{S_{b,\text{new}}}{S_{b,\text{old}}} \left(\frac{V_{b,\text{old}}}{V_{b,\text{new}}} \right)^2$$

Use old and new voltage bases from the same physical side.

$$\text{At fixed } V_b : \quad Z_{\text{p.u.,new}} = Z_{\text{p.u.,old}} \frac{S_{b,\text{new}}}{S_{b,\text{old}}}.$$

A 50% increase in power base at fixed voltage changes 0.040 pu to 0.060 pu.

Example 5B: changing the power base

The same 30 kVA, 1500/150 V transformer has

$$Z_{\text{p.u.,old}} = 0.0200 + j0.045826 \quad \text{on a 30 kVA base.}$$

Choose a new power base of 45 kVA. Keep the voltage bases at 1500 V on H and 150 V on L.

1. Find the new H-side impedance base and per-unit impedance.
2. Reconstruct the H-side impedance in ohms.
3. Does the transformer's physical kVA rating change?

Solution 5B: new base and physical check

$$Z_{b,H,new} = \frac{1500^2}{45,000} = 50.0 \, \Omega$$

$$Z_{p.u.,new} = (0.0200 + j0.045826) \frac{45}{30}$$

$$Z_{p.u.,new} = 0.0300 + j0.0687$$

Using unrounded values to recover physical impedance,

$$(0.0300 + j0.068739)(50.0) = 1.50 + j3.43695 \, \Omega.$$

The physical impedance and 30 kVA nameplate rating stay unchanged. Only the numerical bases change.

PART 6

Planning: screening cost and capacity

Compare annual costs and calculate the capacity requirement

Optional practice

Planning: screening cost and capacity

Optional practice: screening cost and reserve margin

Annual cost per unit of installed capacity is

$$c_i(T) = c_i^{\text{fix}} + c_i^{\text{var}}T \quad [\$/\text{MW-year}].$$

With annualized fixed costs, the crossover satisfies

$$T^* = \frac{c_2^{\text{fix}} - c_1^{\text{fix}}}{c_1^{\text{var}} - c_2^{\text{var}}} \quad [\text{h/year}].$$

Required capacity with reserve margin r and full capacity credit is

$$P_{\text{required}} = (1 + r)P_{\text{peak}}.$$

Compare cost at a specified operating duration. Use peak demand to calculate required capacity.

Example 6: cost and required capacity

Annual fixed costs are already given.

Technology	Fixed (\$/MW-year)	Variable (\$/MWh)
GT	75,000	85
CC	195,000	45

1. Find the crossover and the cheaper choice at 2000 h/year.
2. For a separate 160 MW peak and 15% reserve margin, find required capacity. All capacity receives full credit.

Solution 6: screening and capacity

$$75,000 + 85T = 195,000 + 45T$$

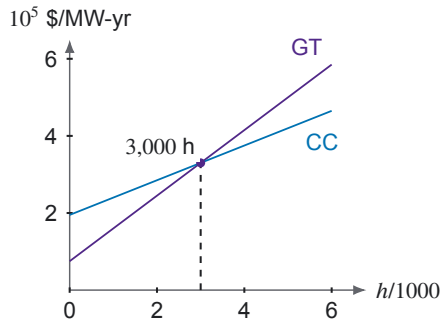
$$T^* = 3000 \text{ h/year}$$

At 2000 h/year, costs are

$$c_{GT} = 245,000, \quad c_{CC} = 285,000$$

in \$/MW-year. GT is cheaper.

$$P_{\text{required}} = 1.15(160) = 184 \text{ MW}$$



The lower cost line identifies the cheaper technology at a stated operating duration.